How-To Library

Informatica

# Implementing Data Masking and Data Subset with Sequential or VSAM Sources

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# Abstract

You can use PowerExchange, PowerCenter, and Persistent Data Masking and Data Subset to perform data masking and data subset with sequential or VSAM sources on z/OS. This article describes the procedures for installing and configuring PowerExchange and performing data masking and data subset with these sources.

# Supported Versions

- PowerCenter 9.5.1, 9.5.2
- PowerExchange 9.5.1, 9.5.2
- Persistent Data Masking and Data Subset 9.5.1, 9.5.2

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# Overview

You can perform data masking and data subset operations on sequential or VSAM sources on z/OS.

To implement data masking or data subset with a sequential or VSAM source, define a PowerExchange data map for the source. After you create a data map, import the source definition into PowerCenter. In the TDM Workbench, you then define PowerExchange NRDB batch connections, include the source in a project, create a plan for data masking and data subset, and generate and run a workflow to implement the plan.

# Implementation Scenario

You are designing a data masking and data subset application that uses PowerExchange to access sequential or VSAM sources on z/OS.

In this scenario, you have already installed and configured PowerCenter and Persistent Data Masking and Data Subset on the appropriate machines. The implementation includes the following components:

- A z/OS machine with sequential or VSAM data sources.

  On this machine, you will install PowerExchange 9.5.1.
- Linux, UNIX, or Windows machine with the Informatica domain and the PowerCenter Integration Service installed. Optionally, this machine also has Informatica Data Explorer components installed.

  On this machine, you will install PowerExchange 9.5.1.
- Windows machine with the PowerCenter Client installed.

  On this machine, you will install PowerExchange 9.5.1, including the PowerExchange Navigator. You will run the PowerExchange Navigator on this system to create data maps. You will run the Workflow Manager to define PowerExchange connections and data sources and edit session properties.
- Windows or UNIX machine with the PowerCenter repository 9.5.1 and TDM Server 9.5.1 or 9.5.2 installed.
- Web browser host from which you log in to the TDM Workbench.

**Note:** You can install PowerCenter on an LPAR running zLinux and use HiperSockets for communications between PowerCenter on one LPAR and the PowerExchange Listener on the LPAR that contains the data source. In this configuration, you must install PowerExchange both on the LPAR with PowerCenter and on the LPAR with the PowerExchange Listener.

# Task Flow

To use Persistent Data Masking and Data Subset with sequential or VSAM sources, complete the following steps:

1. Install and configure PowerExchange on z/OS.
2. Install and configure PowerExchange on the PowerCenter services machine.
3. Install and configure PowerExchange on the PowerCenter Client machine.
4. Define data maps for sequential or VSAM sources and targets on z/OS.
5. Import PowerExchange source definitions into PowerCenter.
6. Log in to the TDM Workbench.
7. Define PowerExchange NRDB Batch connections.
8. Import source and target connections into the TDM repository.
9. Define data masking rules and policies.
10. Create a project to mask specific data sources and to define a subset of the data.
11. Optionally, create and run profiles and import them into the TDM repository.
12. Define entities and groups in data subset.
13. Assign policies and rules to the source data for data masking.
14. Create a plan to run the data subset and masking operations.
15. Generate a workflow from the plan.
16. Run the workflow and monitor the job.

# Step 1. Install and Configure PowerExchange on z/OS

Install PowerExchange on the z/OS system where the IMS unload data file source is located. Then edit the DBMOVER configuration file, configure the PowerExchange Listener JCL, and start the PowerExchange Listener.

## *Install PowerExchange on z/OS*

To install PowerExchange on the z/OS system where the source is located, run the z/OS Installation Assistant. For detailed instructions, see the *PowerExchange Installation and Upgrade Guide*. This guide also gives details about access to the z/OS databases and files.

## *Edit the DBMOVER Configuration Member on the z/OS System*

Edit the DBMOVER configuration member on the z/OS machine.

Ensure that the DBMOVER member contains the following statements:

- LISTENER statement. The LISTENER statement defines the TCP/IP port on which a named PowerExchange Listener process listens for work requests. The default DBMOVER member includes the required LISTENER statement.
- IMSID statement. This statement defines the IMS subsystem information that PowerExchange uses for IMS bulk data movement operations.

For more information about the DBMOVER configuration member, see the *PowerExchange Reference Manual*.

## Configure the PowerExchange Listener JCL on z/OS

Configure the PowerExchange Listener JCL before starting the PowerExchange Listener the first time.

PowerExchange provides PowerExchange Listener JCL in the following members of the RUNLIB library:

- PWXLSTNR runs PowerExchange as a started task. When you ran the XIZZZ998 job during installation, the PWXLSTNR member was copied to the PROCLIB library.
- STARTLST runs PowerExchange as a batch job.

To configure the PowerExchange Listener JCL:

1. Verify that the STEPLIB DD concatenation in the PowerExchange Listener JCL includes the load libraries for your source DBMSs.

   Usually, the PowerExchange LOADLIB library occurs before the data-source load libraries in the concatenation.

   The STEPLIB concatenation in the STARTLST member includes the load libraries for all data sources in a proper order.
2. Verify that the PARM statement specifies the node name that is defined in the LISTENER statement of the DBMOVER configuration file.
3. In the REGION parameter of the EXEC statement, enter a region size. The region size indicates the amount of storage to allocate for running the PowerExchange Listener.

   The sample JCL specifies a region size of 384 MB in the EXEC statement.

   Informatica recommends a region size of 0M, which indicates that z/OS gives the job all the storage available below and above 16 megabytes. For important considerations about specifying a region size of 0M, see the *z/OS MVS JCL Reference*.

   If you omit the REGION parameter, the system default applies.
4. Verify that the TIME parameter in the EXEC statement is set to NOLIMIT. The NOLIMIT setting enables the PowerExchange Listener job to run without a time limit and prevents timeout-related abends that might occur with abend code S322.
5. If the SECURITY statement in the DBMOVER configuration file is set to 1 or 2, APF-authorize the PowerExchange LOADLIB library and any other load libraries in the STEPLIB DD concatenation of the PowerExchange Listener JCL.

For more information, see the *PowerExchange Bulk Data Movement Guide*.

## Start the PowerExchange Listener

To start the PowerExchange Listener, issue the MVS START command with the name of the started task.

Use the following syntax:

```
START listener_task_name
```

You can also run the PowerExchange Listener as a batch job. However, because it is a long-running task, using an MVS started task is more appropriate.

# Step 2. Install and Configure PowerExchange on the PowerCenter Services Machine

Install and configure PowerExchange on the machine where the PowerCenter Integration Service runs. In this scenario, all PowerCenter services, including the PowerCenter Integration Service, reside on the same machine.

## *Install PowerExchange on the PowerCenter Services Machine*

Install PowerExchange on the PowerCenter services machine.

Use the PowerExchange installation instructions for the operating system type and bit-level. If PowerCenter services run on a 64-bit machine, use the PowerExchange 64-bit installation executable or tar file. If PowerCenter services run on a 32-bit machine, use the PowerExchange 32-bit executable or tar file.

For detailed installation instructions, see the *PowerExchange Installation and Upgrade Guide*.

## *Edit the DBMOVER Configuration File on the PowerCenter Services Machine*

Edit the DBMOVER configuration file on the PowerCenter services machine.

Include a NODE statement to define the TCP/IP host name and port that PowerExchange uses to contact the PowerExchange Listener on the z/OS system.

**Syntax:**

```
NODE=({node_name|node1}
      ,TCPIP
      ,host_name
     ,{port|2480}
    [,{send_bufsize|65536}]
    [,{receive_bufsize|65536}]
    [,{send_size|4096}]
    [,{receive_size|4096}]
    [,receive_timeout]
    [,{SSL|ZOSSSL}]
    [,service_name]
)
```

For more information, see the *PowerExchange Reference Manual*.

# Step 3. Install and Configure PowerExchange on the PowerCenter Client Machine

Install and configure PowerExchange on the machine where you installed the PowerCenter Client. You will run the PowerExchange Navigator on this machine.

## *Install PowerExchange on the PowerCenter Client Machine*

Install PowerExchange on the machine where the PowerCenter Client is installed.

Run the PowerExchange 32-bit executable file. You must use the Windows 32-bit executable to install the PowerExchange Navigator, even on a Windows 64-bit machine.

For detailed installation instructions, see the *PowerExchange Installation and Upgrade Guide*.

## *Edit the DBMOVER Configuration File on the PowerCenter Client Machine*

Edit the DBMOVER configuration file on the PowerCenter Client machine.

Include a NODE statement to define the TCP/IP host name and port that PowerExchange uses to contact the PowerExchange Listener on the z/OS system.

**Syntax:**

```
NODE=({node_name|node1}
      ,TCPIP
      ,host_name
```

```
    ,{port|2480}
   [,{send_bufsize|65536}]
   [,{receive_bufsize|65536}]
   [,{send_size|4096}]
   [,{receive_size|4096}]
   [,receive_timeout]
   [,{SSL|ZOSSSL}]
   [,service_name]
)
```

For more information, see the *PowerExchange Reference Manual*.

## Step 4. Define PowerExchange Data Maps for z/OS Sources and Targets

Create a PowerExchange data map for each sequential or VSAM source.

Each data map must be available to the source and corresponding target. If a target is on the same z/OS system as the source, you can use the same data map for the source and target. Otherwise, send the data map to both the source and target systems.

1. Start the PowerExchange Navigator.
2. On the **Resources** tab in the **Resource Explorer**, click **Add** > **Data Map**.
3. In the **Name** dialog box, enter the following information:

| Option | Description |
|---|---|
| Schema Name | Enter a user-defined schema name.<br>PowerExchange uses this name as the first part of the data map name, as follows:<br>*schema_name.data_map_name*<br>The name must consist of alphanumeric characters and begin with a letter.<br>Maximum length is 10 characters. |
| Data Map Name | Enter a user-defined map name.<br>PowerExchange uses this name as the second part of the data map name, as follows:<br>*schema_name.data_map_name*<br>The name must consist of alphanumeric characters and begin with a letter.<br>Maximum length is 10 characters. |
| Access Method | Select one of the following access methods:<br>- **ESDS**. VSAM entry sequenced data set (ESDS) on z/OS.<br>- **KSDS**. VSAM key sequenced data set (KSDS) on z/OS.<br>- **RRDS**. VSAM relative record data set (RRDS) on z/OS.<br>- **SEQ**. Sequential data set on z/OS. |
| Import Record Definitions | Select this option to import a copybook into the data map.<br>Clear this option to add only the data map.<br>By default, this option is selected. |

4. Enter any additional information that is required for the access method that you selected.
5. If you selected the **Import Records Definitions** option, click **Next**.

   If you cleared this option, click **Finish**.
6. If you selected the **Import Records Definitions** option, provide the required information in the dialog boxes.

7. For each data map, click **File > Send to Remote Node**, and complete the **Data Map Remote Node** dialog box to send the data map to the PowerExchange Listener on the source system.

   In addition, if different Listeners will write to the source and target, send each data map to the Listener on the target system as well.

For more information about creating data maps, see the *PowerExchange Navigator User Guide*.

## Step 5. Import PowerExchange Source Definitions into PowerCenter

When you import source definitions into PowerCenter, they become available for import into the TDM Workbench.

After you connect to a source database from PowerCenter, you can import source metadata into the PowerCenter repository. You must create a folder in the PowerCenter repository to store source definitions and metadata that you import into the PowerCenter repository. When you import metadata from PowerCenter into the TDM repository, the TDM Workbench creates a copy of the PowerCenter folder in the TDM repository.

You do not need to import PowerExchange target definitions into PowerCenter. When you generate a workflow from the TDM Workbench, the required target definitions are generated automatically.

**Note:** Persistent Data Masking and Data Subset 9.5.1 and 9.5.2 do not support importing multiple records at a time from sequential or VSAM data sources with multiple-record data maps. You can include only one record and table per source definition.

1. In the PowerCenter Designer, connect to a PowerCenter repository.
2. Create a folder for the source that you are importing.
3. Click **Sources > Import from PowerExchange**.
4. Enter the following information in the dialog box:

| Attribute | Required/ Optional | Description |
|---|---|---|
| Location | Required | Name of the PowerExchange Listener (node name from PowerExchange dbmover.cfg) on the system on which the database resides. |
| User Name | Required | User name for the database to connect to. |
| Password | Required | Password associated with the user name. |
| Multi-Record Datamaps | Optional | Clear this option. You can import only a single record and table per source definition for data subset and masking operations. |
| Use Sequence Fields | Optional | Clear this option. This option applies only to multi-record data maps. |
| Source Type | Required | Select **SEQ** or **VSAM**. |
| CDC Datamaps | n/a | Clear this check box. |
| Network Timeout | Optional | Number of seconds that elapse during a network send or receive operation before PowerExchange ends the connection and issues a timeout error message. |

| Attribute | Required/ Optional | Description |
|---|---|---|
| Schema | Optional | Enter a schema name to filter the resulting data maps. |
| Map name | Optional | Enter a data map name to filter the resulting data maps. |
| Respect Case | Optional | If this attribute is selected, PowerExchange filters data maps by using the exact case of the Schema and Map name fields. Otherwise, PowerExchange ignores the case when searching for data maps. |
| Selected Datamaps | n/a | Lists the available data maps for the connection, source type, and filter details that you entered. |

5. Optionally, enter a filter to view particular schemas and tables from the database.

   You can also enter a filter condition to display schemas and tables that meet the filter condition. To use a filter condition, enter a schema name in **Schema**, a table name in **Map name**, or enter both values.

6. Click **Connect**.

   The available schemas appear in the **Selected Datamaps** list.

7. Expand the schema to display all the tables for all the data maps in the schema, and select the tables that you want to import. You can import multiple tables from a single data map or from multiple data maps.

8. Click **OK**.

   The source definitions appear. The PowerCenter source is named *mapname_tablename*.

# Step 6. Log In to Test Data Manager

Test Data Manager is a web-based user interface that you use to complete data discovery, data subset, and data masking operations. Open a view in Test Data Manager based on the task that you need to perform.

## *Logging In to Test Data Manager*

To access Test Data Manager, enter the host name and port number of the TDM Server in a web browser.

To log in, enter a user name and password defined in Informatica Administrator.

1. In the address bar of a web browser, enter the Test Data Manager URL.

   - Use the following format if Transport Layer Security is enabled:

     `https://hostname:portnumber/tdm/`

   - Use the following format if Transport Layer Security is not enabled:

     `http://hostname:portnumber/tdm/`

   Where:

   - *hostname* is the host name or IP address of the machine where you installed the TDM Server.
   - *portnumber* is the port number. The default is 6643 if Transport Layer Security is enabled. The default is 6605 if Transport Layer Security is not enabled.

   For example, you might enter the following URL:

   `http://TXW1779:6643/tdm/`

   The **Login** dialog box of Test Data Manager appears.

2. Enter the user name and password.

Select the security domain. If the Informatica domain is configured to use LDAP authentication, the default security domain **Native**.

3. Click **Login**.

   Test Data Manager opens.

To log out of Test Data Manager, click **Logout**.

# Step 7. Define PowerExchange NRDB Batch Connections

Define PowerExchange NRDB batch connections in Test Data Manager.

Define separate connections for the source and target, even if you connect to them using the same PowerExchange Listener.

When you generate a workflow, you can select these source and target connections.

Before you define source and target connections, verify that the connection to the PowerCenter repository is defined in Test Data Manager.

**Note:** Instead of defining the connections in Test Data Manager, you can define the connections in PowerCenter and import them into the TDM repository.

## *PowerExchange NRDB Batch Connections*

You can create a PowerExchange NRDB Batch connection in Test Data Manager to perform data subset and data masking operations.

The following table describes the database connection properties for a PowerExchange NRDB Batch database:

| Property | Description |
|---|---|
| Name | Required. Name of the connection. The name is not case sensitive and must be unique within the domain. It cannot exceed 128 characters, start with a number, contain spaces, or contain the following special characters:<br>~ ` ! $ % ^ & * ( ) - + = { [ } ] \| \ : ; " ' < , > . ? / |
| Connection Type | Required. The connection type. Select PWX NRDB Batch. |
| Description | The description of the connection. The description cannot exceed 255 characters. |
| User Name | Required. The database user name. |
| Password | Required. The password for the database user name. |
| Owner | The owner of the connection. Default is the user who creates the connection. You can change the owner of the connection. |
| Code Page | The code page that the Integration Service uses to extract data from the source database.<br>**Note:** In Unicode mode, PWXPC sets the code page with this value overriding any code page specification in the PowerExchange configuration file. |

| Property | Description |
|---|---|
| Array Size | Storage array size, in number of records, that is used for partitioned or multithreaded sessions. For partitioned sessions, this array size is shared across the partitions. Valid values are from 1 through 100000.<br>Default is 25.<br>**Note:** To tune partitioned sessions, particularly when the Write Mode attribute specifies Confirm Write On, increase the array size. |
| Write Mode | Mode in which the Integration Service sends data to the PowerExchange Listener. Configure one of the following write modes:<br>- Confirm Write On. Sends data to the PowerExchange Listener and waits for a response before sending more data. Select if error recovery is a priority. This option might decrease performance.<br>- Confirm Write Off. Sends data to the PowerExchange Listener without waiting for a response. Use this option when you can reload the target table if an error occurs.<br>- Asynchronous With Fault Tolerance. Sends data to the PowerExchange Listener without waiting for a response. This option also provides the ability to detect errors. This provides the speed of Confirm Write Off with the data integrity of Confirm Write On.<br>Default is Confirm Write On. |
| Off Load Processing | Controls whether to use offload processing. Offload processing transfers PowerExchange bulk data processing from the source system to the Integration Service machine. Select one of the following values:<br>- No. Do not use offload processing.<br>- Auto. PowerExchange determines whether to use offload processing.<br>- Filter After. Offloads the bulk data processing to the PowerCenter Integration Service machine, including the filtering of data.<br>- Filter Before. Offloads processing to the PowerCenter Integration Service machine but continues to filter data on the source system.<br>If you select Filter Before for an IMS unload file data source, PowerExchange changes the selection to Filter After.<br>Default is No. |
| Worker Threads | The number of threads that PowerExchange uses on the Integration Service machine to process bulk data. For optimal performance, this value should not exceed the number of installed or available processors on the Integration Service machine. Valid values are from 1 through 64.<br>Default is 0, which disables multithreading. If you use reader or writer pipeline partitioning, accept the default value of 0. You cannot use both multiple worker threads and partitioning. |
| Connection Retry Period | Number of seconds the Integration Service attempts to reconnect to the data source or target if the connection fails. If the Integration Service cannot connect to the database in the retry period, the operation fails.<br>Default is 0. |
| Compression | Select to compress source data during the PowerCenter session.<br>Default is disabled. |
| Encryption Type | The type of encryption that the Integration Service uses. Select one of the following values:<br>- None.<br>- RC2.<br>- DES.<br>Default is None. |

| Property | Description |
|---|---|
| Encryption Level | The level of encryption that the Integration Service uses. If you select RC2 or DES for Encryption Type, select one of the following values to indicate the encryption level.<br>- 1. Uses a 56-bit encryption key for DES and RC2.<br>- 2. Uses a 168-bit triple encryption key for DES. Uses a 64-bit encryption key for RC2.<br>- 3. Uses a 168-bit triple encryption key for DES. Uses a 128-bit encryption key for RC2.<br>Ignored if you do not select an encryption type.<br>Default is 1. |
| Interpret As Rows | Interprets the pacing size as rows or kilobytes. Select to represent the pacing size in number of rows. If you clear this option, the pacing size represents kilobytes.<br>Default is disabled. |
| Convert Character Data As String | Convert character fields to string fields so that embedded nulls in data are processed as spaces.<br>Default is disabled. |
| PWX Override | PowerExchange connection override values, separated by semicolons. You can enter any of the following overrides:<br>`TCPIP_OP_TIMEOUT=network_operation_timeout;`<br>`TCPIP_CON_TIMEOUT=connection_timeout;`<br>`WRT_ERROR_HANDLING=Y`<br>If you use write partitioning and set the Write Mode attribute to Confirm Write On, you can specify `WRT_ERROR_HANDLIN=Y` to use an alternative method of error handling that is more efficient than the default method when the input data contains many errors. You can also specify this statement in the dbmover.cfg configuration file. |
| Pacing Size | Amount of data that the source system can pass to the PowerExchange Listener. Configure the pacing size if an external application, database, or the Integration Service node is a bottleneck. The lower the value, the faster the performance. Enter 0 for maximum performance.<br>Default is 0. |
| Location | Required. Location of the PowerExchange Listener node. The location is defined in the first parameter of the NODE statement in the PowerExchange dbmover.cfg configuration file. |
| Get PowerExchange Log Entries | Specifies whether PWXPC writes PowerExchange messages related to a CDC session in the PowerCenter session log.<br>Default is disabled. |

## *Creating a Connection*

You can create a source or target connection in the TDM Workbench.

1. In the **Administrator | Connections** view, select New Connection from the **Actions** menu.

   A tab opens to display the new connection properties.
2. Select **PWX NRDB Batch** for the connection type and define the connection name, description, and user information.
3. Click **Next**.
4. Enter the connection properties.
5. Click **Test Connection** to test the connection.
6. Click **Finish** to save the connection.

   The connection is visible in the **Administrator | Connections** view.

# Step 8. Define Data Masking Rules and Policies

Use data masking to replace source data in sensitive columns with realistic test data for non-production environments. When you create data masking rules, you define the logic used to replace sensitive data. To specify the sensitive columns that you want to mask, assign data masking rules to source columns and policies.

A policy is a collection of data masking rules. When you assign rules to policies, you can assign multiple source columns to the data masking rules. You can also assign a rule directly to a source column. You can assign data masking rules based on the datatype of the source columns.

Complete the following high-level steps to define policies:

1. Create data masking rules.
2. Define data domains to describe which columns can receive the same masking rules.
3. Assign the data masking rules to the data domains.
4. Create a policy and assign the data domains to the policy.

## *Creating a Standard Masking Rule*

Create a rule to define a masking technique, the datatype to mask, and masking parameters that define how to apply the technique.

1. To access the **Policies** view, click **Policies**.
2. Click **Actions** > **New** > **Masking Rule**.

   The **Rule Wizard** appears.
3. Enter a name and optional description for the rule.
4. Select the datatype of the column to apply the masking rule to.
5. Select the Standard masking rule.
6. To enable users to override masking parameters for a rule, select the **Override Allowed** option.
7. Click **Next**.

   **Note:** The **Masking Parameters** dialog box changes based on the **Masking Technique** you select.
8. Enter the masking parameters.
9. Enter the exception handling options. Configure how to handle null or empty spaces. Configure whether to continue processing on error.
10. Click **Finish**.

## *Creating an Advanced Masking Rule*

Create an advanced masking rule to combine more than one masking technique or to mask multiple columns.

1. To access the **Policies** view, click **Policies**.
2. Click **Actions** > **New** > **Masking Rule**.

   The **Rule Wizard** appears.
3. Enter a name and optional description for the rule.
4. Select the Advanced masking rule.
5. Click **Next**.
6. In the **Input Columns** section, click **Add New**.

The **Add Column** dialog box appears.

7. Enter the column properties, such as the name, datatype, precision, and scale. Select whether the column is mandatory to assign to a data source in all projects.

   You cannot enter masking properties for input columns.
8. Click **OK**.
9. To enter more input columns, click **Create Input Column** in the **New Masking Rule** dialog box.
10. In the **Variable Columns** section, click **Add New**.
11. Enter an expression, a masking rule, or a dependent column for each variable column that you define. If you apply a masking rule, configure the input column to create the variable column from.
12. Click **OK**.
13. In the **Output Columns** section, click **Add New**.
14. Enter an expression, a masking rule, or a dependent column for the output column.
15. Click **OK**.

## *Creating a Data Domain*

When you create a data domain, you can enter regular expressions that describe the data that you want to include in the data domain. You can also enter regular expressions that describe the names of database columns to include.

1. To access the policies, click **Policies**.

   The **Policies** view shows a list of the policies, data domains, and rules in the TDM repository.
2. Click **Actions** > **New** > **Data Domain**.
3. Enter the name, sensitivity level, and description for the data domain. Click **Next**.
4. Click **Next**.
5. Optionally, enter a regular expression to filter columns by data pattern.
6. To add more expressions for data patterns, click the **+** icon.
7. To add regular expressions that filter columns by column name, click **Next**. Or, click **Finish** to skip entering any more data domain information.

   You can add multiple expressions.
8. Enter regular expressions to filter columns by column name.
9. Click **OK**.
10. Click **Finish**.

## *Creating a Policy*

Create a policy in the **Policies** view. You can create the policy before you create data domains and rules. You can add the data domains and rules to the policy at any time.

1. In the **Policies** view, click **Actions** > **New** > **Policy**.

   The **New Policy** dialog box appears.
2. Enter a name and optional description for the policy, and click **Next**.
3. To add data domains to the policy, click **Add Data Domains**.
4. Select the data domains from the list.

5. Click **Finish**.

   The policy appears in the **Policies** view.

# Step 9. Create a Project

Create a project folder to contain the data discovery, subset, and masking components to apply to a data source.

## *Creating a Project*

1. In Test Data Manager, click **Projects** to access the projects.

   A list of projects appears.
2. Click **Actions** > **New**.
3. In the **Create Project** dialog box, enter project properties. The following table describes the project properties:

| Option | Description |
|---|---|
| Description | The description of the project. |
| Folder | The name of the project folder in the repository. Default is the project name. You can choose another folder in the repository. |
| Owner | The name of the user that owns the folder. The folder owner has all permissions on the folder. The default is the name of the user that created the folder. You can select another user as the folder owner. |

4. Click **OK**.

   The project opens in a new page.

## *Importing Data Sources*

Import data sources from PowerCenter to a project in the Workbench.

1. Click **Projects** to view a list of the projects in the Workbench.
2. Click a project in the list to open it in the Workbench.
3. Click **Actions** > **Import Metadata**.
4. Select to import metadata from the PowerCenter repository.
5. Choose whether you want to review the metadata changes before you import the data sources. The Workbench displays the information before the import runs. You can choose to skip the import.
6. Click **Next**.
7. Select the PowerCenter folder that contains the data source you want to import.
8. Click **Next**.
9. Select the tables to import.
10. Click **Next**.

11. Choose when to import the sources. Choose one of the following options:
    - Import now.

      If you select this option, you also choose whether to run the import in the background.
    - Schedule later.

      Schedule the import to occur at a specific date and time. Click the calendar to select a date. Use the hour and minute sliders to set the time.
12. Click **OK**.

    If the import is running, view the progress of the import job in the **Monitor** view. After the job finishes, access the imported metadata through the **Data Sources** details view.

# Step 10 (Optional). Run Profiles for Data Discovery

You cannot use Test Data Manager to run a profile for data discovery on nonrelational sources. However, you can create and run profiles in Informatica Data Explorer and import them into the TDM repository.

Complete the following high-level steps to perform data discovery:

1. Create profiles in Informatica Data Explorer.
2. Run the profiles in Informatica Data Explorer.
3. Export the results from Informatica Data Explorer.
4. Open Test Data Manager, and import the results to the TDM repository.

# Step 11. Define Groups in Data Subset

Define groups for a data subset project. After you create a plan, you can add the groups to the plan.

**Note:** You can also add entities and templates to a data subset project. However, these objects are of limited usefulness for nonrelational sources.

## *Creating a Group*

To create a group, select the tables that you want to add to a data subset plan.

1. On the **Define** | **Data Subset** view in the project, click **Actions** > **New** > **Groups**.
2. In the **Create Group** dialog box, enter a name for the group and optional description for the group. Click **Next**.
3. To select one or more tables for the group, click **Add Tables**.
4. Optionally, you can filter the list of tables to search for.
5. Select the required tables and click **OK**.
6. Click **Save**.

# Step 12. Assign Policies and Masking Rules to the Source Data

To implement data masking operations, assign masking rules to columns in a source.

You can assign a rule to a column through a data domain default rule or by choosing a rule. If the column belongs to a data domain, the default rule appears in the column assignment.

## *Assigning a Standard Masking Rule*

You can assign a standard masking rule to a target column. Assign masking rules from data domains or policies to one or more columns in the project source that you want to mask.

1. In the project, click **Define** | **Data Masking** to access the **Data Masking** view.
2. Select a column to assign a masking rule to.
3. If the **Domain** is blank for the column, click the **Policy** column and choose a policy that contains the data masking rule that you want to assign.
4. Click inside the **Rule** column to view the list of available rules.

   The data domain preferred rules appear at the top of the list. The other rules in the policy appear at the bottom of the list.
5. Select a masking rule. If you choose substitution masking rule, you can specify the rule assignment parameters. If you choose mapplet or advanced rule, you can assign the rule columns to the columns in the project.
6. Click **Save** for each column that you update.
7. If you want to update a masking rule assignment, select another masking rule.

   If the masking rule contains assignments, the **Impacted Objects** dialog box appears with the list of affected plans, columns, and data domains.
8. To download the list of affected columns and plans, click **Export**, and save the .csv file.
9. To save the changes, click **Continue**.

   To update the changes in a plan, you must generate and run the plan again.

# Step 13. Create a Plan

Create a plan to run the data subset and data masking operations.

For detailed information about Test Data Management plans and workflows, see the *Informatica Test Data Management User Guide*.

## *Creating a Data Masking and Data Subset Plan*

1. Open a project and click **Execute** to view the project plans.
2. Click **Actions** > **New**.
3. In the **New Plan** dialog box, enter a name and optional description for the plan.
4. To add a data masking operation to the plan, click **Add Masking Components**.
5. Select the policies and rules to add to the plan. Click **Next**.
6. To add a data subset operation to the plan, click **Add Subset Components**.
7. To skip masking a rule, select the check box for the rule and click **Off**.
8. To filter subset components, select the component and choose to enter a basic or advanced expression.
9. To limit the subset results, click **Limit** and choose to limit by percentage, by absolute value, or by defining an interval of rows to create. Click **Next**.
10. To override plan settings for a data source, select the data source and click **Override Plan Settings** and enter the properties.
11. To override data source settings for a table, select the table and click **Override Data Source Settings** and enter the properties.

12. Click **Finish**.

    The plan appears in the project.

# Step 14. Generate a Workflow

Generate a workflow from a plan, and then run the workflow to perform subset and masking operations.

You can generate and start a workflow in a single step.

When you generate a workflow, the PowerCenter Repository Service generates mappings for the workflow. When you start a workflow, the PowerCenter Integration Service performs the data masking and data subset tasks defined in the plan.

You can view the status of generated workflows or the load configuration on the **Monitor** view.

## *Generating a Workflow*

1. In a project click **Execute** to access the plans in the project.
2. Select a plan from the list.
3. Click **Actions** > **Generate Workflow**.
4. Choose whether to run it immediately or to generate the workflow later.

# Step 15. Run the Workflow and Monitor the Job

Start the workflow to run sessions from the PowerCenter mappings. You can view workflow status in the **Monitor** view.

## *Executing a Workflow*

You can start a workflow from the **Plan | Properties** view or the **Plan | Workflow** view.

1. To generate and run a workflow in a single step, in the **Plan | Properties** page, click **Actions** > **Generate and Execute**.
2. You can generate and run a workflow in separate steps. If the plan you selected contains one workflow, click **Actions** > **Execute Workflow** in either view.
3. If the plan you selected contains multiple workflows, click the **Workflow** view.
   a. Select the workflow that you want to start.
   b. Click **Actions** > **Execute Workflow**.
4. Choose the required Integration Service.
5. Choose to run the workflow immediately or schedule it for a later time.
6. Click the **Generate and Execute** button to run the workflow.
7. View the workflow status in the **Workflow Execution** view.

# Author

**Jim Middleton**